

Code Review Report

Who Can Make24 - Real-time Multiplayer Game

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MASALAH KRITIS (Critical Issues)

Isu-isu berikut HARUS diperbaiki sebelum deployment ke production. Ini merupakan security dan stability risks yang serius.

1 CORS Security: Origin "*" Terlalu Permisif

📍 Lokasi:

apps/server/src/index.ts (Lines 18-22)

⚠️ Masalah:

✖ CURRENT CODE:

```
const io = new Server(httpServer, {
  cors: {
    origin: "*", // ⚠️ CRITICAL: Accepts from ANY domain
    methods: ["GET", "POST"],
  },
  transports: ["websocket", "polling"],
});
```

Impact: Konfigurasi CORS origin: "*" memungkinkan **siapa saja dari domain mana pun** untuk terhubung ke server game Anda. Ini membuka celah serius:

- **Cross-Origin Attacks:** Attacker bisa embed game di site mereka dan steal data
- **Bot Abuse:** Mudah untuk membuat bot yang connect tanpa batasan
- **DDoS Risk:** Tidak ada kontrol terhadap source koneksi

✓ Solusi & Contoh Kode:

✓ FIXED CODE:

```
import dotenv from 'dotenv';
dotenv.config();

const allowedOrigins = (
  process.env.ALLOWED_ORIGINS ||
  'http://localhost:5173'
).split(',');

const io = new Server(httpServer, {
  cors: {
    origin: allowedOrigins, // ✓ Whitelist only
    credentials: true,
    methods: ["GET", "POST"],
  },
```

```
    perMessageDeflate: false, // Reduce memory  
  });
```

.env Configuration:

ENVIRONMENT FILE:

```
ALLOWED_ORIGINS=https://yourgame.com,https://www.yourgame.com,http://localhost:517
```

▮ **Alasan Perbaikan:** Melindungi dari unauthorized cross-origin requests, mengurangi attack surface, dan memastikan hanya aplikasi Anda yang bisa konek ke server.

📍 Lokasi:

```
apps/server/src/rooms/roomHandlers.ts (Lines 23-30)
```

⚠️ Masalah:

```
✗ CURRENT CODE: socket.on("room:create", async ({ name, roomName, avatar, mode  
}) => { const host: Player = { ...makePlayer(name, avatar), isHost: true }; //  
✗ Tidak ada validasi sama sekali!
```

Risiko:

- **XSS Risk:** Player bisa submit `<img src=x onerror="alert('xss')">` sebagai nama
- **Data Integrity:** Nama bisa berisi karakter invalid, emoji berlebihan, null bytes
- **Buffer Overflow:** Nama panjang unlimited bisa cause memory issues

✓ Solusi & Contoh Kode:

✓ VALIDATION UTILITIES:

```
// Validate player name  
function validatePlayerName(name: string) {  
  if (!name || typeof name !== 'string') {  
    return { valid: false, error: 'Name required' };  
  }  
  if (name.length < 2 || name.length > 20) {  
    return {  
      valid: false,  
      error: 'Name must be 2-20 characters'  
    };  
  }  
  // Only allow letters, numbers, spaces, underscore  
  if (!/^[\\p{L}\\p{N}\\s_~]+$/u.test(name)) {  
    return { valid: false, error: 'Invalid characters' };  
  }  
  return { valid: true };  
}  
  
// Validate room name  
function validateRoomName(name: string) {
```

```

if (!name || typeof name !== 'string') {
  return { valid: false, error: 'Room name required' };
}
if (name.length < 3 || name.length > 30) {
  return {
    valid: false,
    error: 'Room name must be 3-30 chars'
  };
}
if (!/^[\\p{L}\\p{N}\\s_-]+$/.test(name)) {
  return { valid: false, error: 'Invalid characters' };
}
return { valid: true };
}

// Validate avatar from whitelist
const VALID_AVATARS = [' ', ' ', ' ', 'x', ' ', ' '];
function validateAvatar(avatar: string): boolean {
  return VALID_AVATARS.includes(avatar);
}

// Validate game mode
function validateGameMode(mode: string): boolean {
  return ['casual', 'pvp'].includes(mode);
}

```

✓ HANDLER WITH VALIDATION:

```

socket.on("room:create",
  async (data) => {
    // Validate player name
    const nameValidation = validatePlayerName(data.name);
    if (!nameValidation.valid) {
      socket.emit("room:error", {
        message: nameValidation.error
      });
      return;
    }

    // Validate room name
    const roomValidation = validateRoomName(
      data.roomName
    );
    if (!roomValidation.valid) {

```

```

        socket.emit("room:error", {
            message: roomValidation.error
        });
        return;
    }

    // Validate avatar
    if (!validateAvatar(data.avatar)) {
        socket.emit("room:error", {
            message: 'Invalid avatar'
        });
        return;
    }

    // Validate game mode
    if (!validateGameMode(data.mode)) {
        socket.emit("room:error", {
            message: 'Invalid game mode'
        });
        return;
    }

    // ✔ All inputs are now safe!
    const host: Player = {
        ...makePlayer(data.name, data.avatar),
        isHost: true
    };
    const room = await createRoom(
        data.roomName,
        data.mode,
        data.isPrivate,
        host
    );
}
);

```

▣ **Alasan Perbaikan:** Validasi input adalah first line of defense terhadap XSS, injection, dan data corruption. Melindungi semua pengguna dari malicious input.

3

No Rate Limiting - Vulnerable to Socket Spam & DoS

📍 Lokasi:

apps/server/src/index.ts (Lines 42-47)

⚠️ Masalah:

✖ CURRENT CODE (NO PROTECTION):

```
io.on("connection", (socket) => {  
  console.log(`Player connected: ${socket.id}`);  
  registerRoomHandlers(io, socket);  
  registerGameHandlers(io, socket);  
  // ✖ Zero rate limiting!  
});
```

Risiko: Tidak ada batasan berapa kali player bisa emit event. Akibatnya:

- **Socket Spam:** Player bisa emit game:start 1000x per detik
- **Resource Exhaustion:** Server keteteran process event, memory penuh
- **DoS Attack:** Attacker bisa crash server dengan simple bot

✓ Solusi & Contoh Kode:

✓ RATE LIMITER UTILITY:

```
type RateLimitConfig = {  
  maxEvents: number;  
  windowMs: number;  
};  
  
function createRateLimiter(config: RateLimitConfig) {  
  return {  
    attempts: new Map(),  
  
    isAllowed(socketId: string): boolean {  
      const now = Date.now();  
      const key = socketId;  
  
      if (!this.attempts.has(key)) {  
        this.attempts.set(key, [now]);  
        return true;  
      }
```



```

    }

    const timestamps = this.attempts.get(key)!;
    const recentAttempts = timestamps.filter(
      (t) => now - t < config.windowMs
    );

    if (recentAttempts.length >= config.maxEvents) {
      return false;
    }

    recentAttempts.push(now);
    this.attempts.set(key, recentAttempts);
    return true;
  }
};
}

```

✓ HANDLER IMPLEMENTATION:

```

// Create limiters per event type
const bellRateLimiter = createRateLimiter({
  maxEvents: 1,
  windowMs: 1000 // 1 press per second
});

const chatRateLimiter = createRateLimiter({
  maxEvents: 5,
  windowMs: 1000 // 5 messages per second
});

const joinRateLimiter = createRateLimiter({
  maxEvents: 2,
  windowMs: 5000 // 2 joins per 5 seconds
});

// Use in handlers:
export function registerGameHandlers(
  io: Server,
  socket: Socket
) {
  socket.on("game:bell", async () => {
    if (!bellRateLimiter.isAllowed(socket.id)) {
      socket.emit("game:error", {

```

```
        message: "Too many bell presses"
    });
    return;
}
// Process bell press...
});

socket.on("game:chat", async ({ text }) => {
    if (!chatRateLimiter.isAllowed(socket.id)) {
        socket.emit("game:error", {
            message: "Sending too fast"
        });
        return;
    }
    // Process chat...
});
}
```

▮ **Alasan Perbaikan:** Rate limiting melindungi server dari abuse dan memastikan fair experience untuk semua pemain. Mencegah DoS attacks dan resource exhaustion.

4

Room Code Too Short - Trivial Brute Force

📍 Lokasi:

```
apps/server/src/rooms/roomManager.redis.ts (Lines 10-12)
```

⚠️ Masalah:

❌ INSECURE ROOM CODE (ONLY 4 CHARS):

```
function generateRoomCode(): string {
  // Only 4 characters = 36^4 = 1.67M combinations
  // Brute-forceable in 4-5 hours!
  return Math.random()
    .toString(36)
    .substring(2, 6)
    .toUpperCase();
}
```

Risiko: Room code hanya **4 karakter alphanumeric** = $36^4 = \sim 1.67$ juta kombinasi:

- Attacker bisa try 100 room codes per detik
- Dalam 4-5 jam, semua private rooms bisa di-brute force
- Weak entropy untuk protecting private games

✅ Solusi & Contoh Kode:

✔️ SECURE ROOM CODE GENERATOR:

```
import { randomBytes } from 'crypto';

function generateRoomCode(): string {
  // 8 characters = 36^8 = 2.8 Trillion combinations
  // Brute-force would take millions of years
  const chars = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789';
  const length = 8;
  let code = '';

  const randomValues = new Uint32Array(1);
  for (let i = 0; i < length; i++) {
    crypto.getRandomValues(randomValues);
    code += chars[randomValues[0] % chars.length];
  }
}
```

```
    return code;
}

// Example outputs:
// "A7X3K9M2"
// "Q2P8L5R1"
// "Z4W7J9S6"
```

▮ **Alasan Perbaikan:** Longer codes exponentially increase brute-force effort. 8 chars = ~2.8 triliun kombinasi, making brute-force impractical (impossible dalam praktik).

5

Memory Leak: Timer Intervals Not Always Cleared

📍 Lokasi:

apps/server/src/game/gameHandlers.ts (Lines 7-25)

⚠️ Masalah:

✖ POTENTIAL MEMORY LEAK:

```
function startTimer(io: Server, roomId: string) {
  clearRoomTimer(roomId); // ✓ Clear old

  const interval = setInterval(() => {
    const remaining = tickTimer(roomId);
    const state = getGameState(roomId);

    if (!state) {
      clearRoomTimer(roomId); // ✓ OK
      return;
    }

    if (remaining <= 0) {
      clearRoomTimer(roomId);
      handleTimerExpired(io, roomId);
    }
    // BUT: Many edge cases NOT handled!
  }, 1000);

  setInterval(roomId, interval);
}

// Issues:
// - deleteGameState() called but timer still running
// - Player disconnect not cleaning up timers
// - 100 games × many timers = memory exhausted
```

✓ Solusi & Contoh Kode:

✓ ROBUST TIMER MANAGEMENT:

```
function startGameTimer(
  io: Server,
  roomId: string,
```

```

onTimerExpired: (io: Server, roomId: string) => void,
) {
  // Always clear old timer first
  const existingInterval = timerIntervals.get(roomId);
  if (existingInterval) {
    clearInterval(existingInterval);
    timerIntervals.delete(roomId);
  }

  const interval = setInterval(() => {
    try {
      const state = getGameState(roomId);

      // Stop timer if state is gone
      if (!state) {
        clearInterval(interval);
        timerIntervals.delete(roomId);
        return;
      }

      const remaining = tickTimer(roomId);

      // Only emit on important changes
      if (remaining % 5 === 0 || remaining <= 5) {
        io.to(roomId).emit("game:timer", {
          seconds: remaining,
          startTime: state.startTime,
        });
      }

      // Clean up when timer expires
      if (remaining <= 0) {
        clearInterval(interval);
        timerIntervals.delete(roomId);
        onTimerExpired(io, roomId);
      }
    } catch (error) {
      console.error(`Timer error for ${roomId}:`, error);
      clearInterval(interval);
      timerIntervals.delete(roomId);
    }
  }, 1000);

  timerIntervals.set(roomId, interval);

```

```
// Safety: Auto-cleanup after max duration
setTimeout(() => {
  if (timerIntervals.has(roomId)) {
    const interval = timerIntervals.get(roomId);
    if (interval) clearInterval(interval);
    timerIntervals.delete(roomId);
  }
}, 60 * 60 * 1000); // 1 hour max
}

// Ensure cleanup when game ends
export function deleteGameState(roomId: string): void {
  const interval = timerIntervals.get(roomId);
  if (interval) {
    clearInterval(interval);
    timerIntervals.delete(roomId);
  }
  gameStates.delete(roomId);
}
```

▮ **Alasan Perbaikan:** Mencegah memory leak yang bisa crash server setelah banyak games dimainkan. Critical untuk production stability.

SARAN PERBAIKAN (Suggestions)

Isu-isu berikut merupakan peningkatan kualitas code yang sangat disarankan. Tidak urgent tapi penting untuk maintainability dan performance jangka panjang.

1 Redundant Timer Code - DRY Principle Violation

📍 Lokasi:

apps/server/src/game/gameHandlers.ts (Lines 7-50)

⚠ Masalah:

✖ DUPLICATE CODE (80% SAME!):

```
function startTimer(io, roomId) {
  clearRoomTimer(roomId);
  const interval = setInterval(() => {
    // ... 40 lines of IDENTICAL logic ...
    if (remaining <= 0) {
      handleTimerExpired(io, roomId); // Different
    }
  }, 1000);
}

export function startPvpTimer(io, roomId) {
  clearRoomTimer(roomId);
  const interval = setInterval(() => {
    // ... SAME 40 lines again ...
    if (remaining <= 0) {
      handlePvpTimerExpired(io, roomId); // Different
    }
  }, 1000);
}
```

✓ Solusi - Extract Generic Timer:

✓ DRY REFACTORED (SINGLE FUNCTION):

```
function startGameTimer(
  io: Server,
  roomId: string,
  onTimerExpired: (io: Server, roomId: string) => void,
) {
  clearRoomTimer(roomId);

  const interval = setInterval(() => {
    const remaining = tickTimer(roomId);
    const state = getGameState(roomId);
```

```

    if (!state) {
        clearRoomTimer(roomId);
        return;
    }

    if (remaining % 5 === 0 || remaining <= 5) {
        io.to(roomId).emit("game:timer", {
            seconds: remaining,
            startTime: state.startTime,
        });
    }

    if (remaining <= 0) {
        clearRoomTimer(roomId);
        // ✔ Flexible callback!
        onTimerExpired(io, roomId);
    }
}, 1000);

setTimerInterval(roomId, interval);
}

// Usage:
startGameTimer(io, roomId, handleTimerExpired);
startGameTimer(io, roomId, handlePvpTimerExpired);

```

▮ **Alasan:** Mengurangi boilerplate 50%, mempermudah maintenance, single source of truth untuk timer logic.

📍 Lokasi:

```
apps/server/src/rooms/roomManager.redis.ts (Lines 13-40)
```

⚠️ Masalah:

❌ JSON OPERATIONS ON CRITICAL PATH:

```
export async function joinRoom(
  roomId: string,
  player: Player,
  code?: string
) {
  // ❌ Read from Redis (JSON.parse)
  const raw = await redis.get(KEYS.room(roomId));
  const room = JSON.parse(raw);

  room.players.push(player);

  // ❌ Write to Redis (JSON.stringify)
  await redis.set(KEYS.room(room.id),
    JSON.stringify(room)
  );
  await redis.set(KEYS.playerRoom(player.id),
    room.id
  );
}

// In high-traffic:
// 100 concurrent joins = 200+ JSON operations
// Each operation blocks execution
// = severe bottleneck!
```

✅ Solusi - Implement Caching Layer:

✅ OPTIMIZED WITH IN-MEMORY CACHE:

```
class RoomCache {
  private cache = new Map();
  private dirty = new Set();

  async get(roomId: string): Promise {
```

```
// ✓ Check memory cache first
if (this.cache.has(roomId)) {
    return this.cache.get(roomId)!;
}

// Fetch from Redis if not cached
const raw = await redis.get(
    KEYS.room(roomId)
);
if (!raw) return null;

// Parse once and cache
const room = JSON.parse(raw);
this.cache.set(roomId, room);
return room;
}

set(room: Room): void {
    // Keep in-memory version
    this.cache.set(room.id, room);
    // Mark for batch save
    this.dirty.add(room.id);
}

async flush(): Promise {
    // ✓ Batch save all dirty rooms at once
    const updates: Array<[string, string]> = [];

    for (const roomId of this.dirty) {
        const room = this.cache.get(roomId);
        if (room) {
            updates.push([
                KEYS.room(roomId),
                JSON.stringify(room),
            ]);
        }
    }

    // Single Redis call for all updates
    if (updates.length > 0) {
        await redis.mset(updates.flat());
    }

    this.dirty.clear();
}
```

```

}

// Flush every 5 seconds
const roomCache = new RoomCache();
setInterval(() => roomCache.flush(), 5000);

// Usage in handlers:
export async function joinRoom(
  roomId: string,
  player: Player
) {
  const room = await roomCache.get(roomId);
  if (!room) {
    return { success: false, error: "Not found" };
  }

  room.players.push(player);
  roomCache.set(room); // ✓ Mark dirty

  return { success: true, room };
}

```

▮ **Alasan:** Batch operations lebih efficient, mengurangi JSON overhead, better performance at scale.

3

No Input Type Validation on GameState

📍 Lokasi:

apps/server/src/game/gameHandlers.ts (Lines 115-140)

⚠ Masalah:

✖ NO TYPE GUARDS:

```
socket.on("game:bell", async () => {
  const state = getGameState(roomId);

  // ✖ Assuming state is always valid!
  if (state.bellPressers.includes(playerId)) { }
  if (state.phase !== "playing") { }

  // If state is null or wrong structure
  // → Runtime crash, user gets blank screen
});
```

✓ Solusi - Add Type Guards:

✓ TYPE GUARD IMPLEMENTATION:

```
// Define strict type guard
function isValidGameState(
  state: any
): state is GameState {
  return (
    state &&
    typeof state === 'object' &&
    Array.isArray(state.bellPressers) &&
    Array.isArray(state.candidates) &&
    typeof state.phase === 'string' &&
    typeof state.timer === 'number' &&
    Array.isArray(state.proofs) &&
    typeof state.scores === 'object' &&
    Array.isArray(state.roundScores) &&
    Array.isArray(state.pointingTargets)
  );
}

socket.on("game:bell", async () => {
```

```
const state = getGameState(roomId);

// ✔ Validate before use
if (!state || !isValidGameState(state)) {
  socket.emit("game:error", {
    message: "Invalid game state"
  });
  return;
}

// ✔ Now TypeScript knows state is valid
if (state.phase !== "playing") {
  socket.emit("game:error", {
    message: "Not in playing phase"
  });
  return;
}

// ✔ Safe to use all properties
pressBell(roomId, socket.id, room!.players.length);
});
```

▮ **Alasan:** Defensive programming mencegah runtime errors. TypeScript akan help catch bugs at compile time.

4

Scattered Magic Numbers - Use Constants

📍 Lokasi:

Multiple files (gameHandlers.ts, gameManager.ts, WaitingRoom.tsx)

⚠ Masalah:

✖ MAGIC NUMBERS SCATTERED EVERYWHERE:

```
// gameHandlers.ts
setTimeout(() => {
  io.to(roomId).emit("game:round-start", {});
}, 3000); // ✖ Apa 3000? Round delay?

// gameManager.ts
state.totalRounds = Math.floor(deck.length / 4) + 1;
//                                     ↑
//                                     Why +1?

// WaitingRoom.tsx
setTimeout(() => setCopied(false), 2500);
//                                     ↑
//                                     Toast duration?

// Hard to understand, hard to maintain, hard to refactor
```

✓ Solusi - Centralized Constants:

✓ CONSTANTS FILES:

```
// constants/gameTimings.ts
export const GAME_TIMINGS = {
  // Round delays
  ROUND_RESULT_DELAY_MS: 3000,
  ROUND_START_DELAY_MS: 2000,

  // Timer settings
  TIMER_EMIT_INTERVAL_MS: 1000,
  TIMER_BATCH_EMIT_THRESHOLD: 5,

  // UI notifications
  TOAST_NOTIFICATION_DURATION_MS: 2500,
  COPY_SUCCESS_DURATION_MS: 2500,
```



```

// Safety
AUTO_CLEANUP_TIMEOUT_MS: 60 * 60 * 1000, // 1 hour
} as const;

// constants/gameRules.ts
export const GAME_RULES = {
  // Deck composition
  CARDS_PER_ROUND: 4,
  DECK_SUITS: 4,
  DECK_VALUES_CASUAL: 10,
  DECK_VALUES_WILD: 13,
  CARDS_IN_DECK: 52, // or 52 * 13/10
} as const;

// constants/gamePhases.ts
export const GAME_PHASES = {
  PLAYING: 'playing',
  POINTING: 'pointing',
  PROOF: 'proof',
  RESULT: 'result',
  PVP: 'pvp',
  FINISHED: 'finished',
} as const;

```

✓ USAGE EXAMPLES:

```

// gameHandlers.ts
setTimeout(() => {
  io.to(roomId).emit("game:round-start", {});
}, GAME_TIMINGS.ROUND_RESULT_DELAY_MS); // ✓ Clear!

// gameManager.ts
const totalRounds = Math.floor(
  deck.length / GAME_RULES.CARDS_PER_ROUND
) + 1; // ✓ Self-documenting!

// WaitingRoom.tsx
setTimeout(
  () => setCopied(false),
  GAME_TIMINGS.TOAST_NOTIFICATION_DURATION_MS
); // ✓ Obvious intent!

```

▮ **Alasan:** Self-documenting code, easier refactoring, maintainability tinggi. Single source of truth untuk constants.

📍 Lokasi:

`apps/client/src/context/GameContext.tsx (Lines 169-250)`

⚠ Masalah:

❌ CALLBACK HELL - 500+ LINES IN ONE USEEFFECT:

```
export function GameProvider({ children }) {
  const [gameState, setGameState] = useState(null);
  const [timer, setTimer] = useState(0);
  const [phase, setPhase] = useState(null);

  useEffect(() => {
    const onGameStarted = () => {
      // 30 lines...
      setGameState({ ...EMPTY_GAME_STATE });
    };

    const onRoundStart = (data) => {
      // 20 lines...
      setGameState((prev) => ({
        ...prev,
        currentCards: data.cards,
      }));
    };

    const onPhaseChanged = (data) => {
      // 15 lines...
      setPhase(data.phase);
    };

    const onTimer = (data) => {
      // 10 lines...
      setTimer(data.seconds);
    };

    // ... 15+ more handlers ...

    socket.on("game:started", onGameStarted);
    socket.on("game:round-start", onRoundStart);
    // ... cleanup ...
  });
}
```

```

return () => {
  socket.off("game:started", onGameStarted);
  socket.off("game:round-start", onRoundStart);
  // ... cleanup ...
};
}, []);

// Result: 500+ lines, hard to navigate, hard to test
}

```

✓ Solusi - Extract Custom Hooks:

✓ CUSTOM HOOK REFACTORING:

```

// hooks/useGameSocket.ts
export function useGameSocket() {
  const [gameState, setGameState] = useState(null);
  const [timer, setTimer] = useState(0);
  const [phase, setPhase] = useState(null);

  useEffect(() => {
    // ✓ Each handler is small and focused
    const handlers = {
      onGameStarted: () => {
        setGameState({ ...EMPTY_GAME_STATE });
      },
      onRoundStart: (data: RoundStartPayload) => {
        setGameState((prev) =>
          prev ? { ...prev, currentCards: data.cards } : null
        );
      },
      onPhaseChanged: (data: PhaseChangedPayload) => {
        setPhase(data.phase);
        setGameState((prev) =>
          prev ? { ...prev, phase: data.phase } : null
        );
      },
      onTimer: (data: { seconds: number }) => {
        setTimer(data.seconds);
      },
    };
  });

  // Attach listeners
  socket.on("game:started", handlers.onGameStarted);

```

```

socket.on("game:round-start", handlers.onRoundStart);
socket.on("game:phase-changed", handlers.onPhaseChanged);
socket.on("game:timer", handlers.onTimer);

// Cleanup
return () => {
  socket.off("game:started", handlers.onGameStarted);
  socket.off("game:round-start", handlers.onRoundStart);
  socket.off("game:phase-changed", handlers.onPhaseChanged);
  socket.off("game:timer", handlers.onTimer);
};
}, []);

return { gameState, timer, phase };
}

// GameProvider now becomes much cleaner:
export function GameProvider({ children }) {
  const { gameState, timer, phase } = useGameSocket();

  return (

    {children}

  );
}

```

▮ **Alasan:** Improve readability, testability, maintainability. Each hook punya single responsibility.

📍 Lokasi:

```
apps/client/src/App.tsx (Assumed)
```

⚠️ Masalah:

❌ NO ERROR BOUNDARIES = SILENT CRASHES:

```
export function App() {  
  return (  
  
    {/* routes */  
  
  );  
}
```

// Problem:
// If GameContext throws error
// → Entire UI becomes blank screen
// → Error not tracked
// → User has no idea what happened
// → Debugging is nightmare

✅ Solusi - Create Error Boundary:

✓ ERROR BOUNDARY COMPONENT:

```
import React, { ReactNode } from 'react';  
  
interface ErrorBoundaryProps {  
  children: ReactNode;  
  fallback?: (error: Error) => ReactNode;  
}  
  
interface ErrorBoundaryState {  
  error: Error | null;  
}  
  
export class ErrorBoundary extends React.Component<  
  ErrorBoundaryProps,
```

```
ErrorBoundaryState
> {
  constructor(props: ErrorBoundaryProps) {
    super(props);
    this.state = { error: null };
  }

  static getDerivedStateFromError(
    error: Error
  ): ErrorBoundaryState {
    return { error };
  }

  componentDidCatch(
    error: Error,
    errorInfo: React.ErrorInfo
  ) {
    // Log to Sentry, LogRocket, Rollbar, etc
    console.error('Boundary caught error:', error);
    logErrorToService(error, errorInfo);
  }

  render() {
    if (this.state.error) {
      return (
```

Oops! Something went wrong

```
{this.state.error.message}
```

```
        window.location.reload()
      }
      className="px-6 py-2 bg-blue-500 text-white
        rounded hover:bg-blue-600"
    >
      Reload Page
    </button>
  )
}
```

```
    );
  }

  return this.props.children;
}

// Usage in App.tsx:
export function App() {
  return (

    { /* routes */ }

  );
}
```

▮ **Alasan:** Gracefully handle render errors. User tidak melihat blank screen, bisa reload. Error dapat di-track dan di-fix.

7 Batch Room State Updates to Redis

📍 Lokasi:

```
apps/server/src/rooms/roomManager.redis.ts
```

⚠️ Masalah:

```
❌ MULTIPLE REDIS CALLS: export async function joinRoom( roomId, player, code )  
  
{ const raw = await redis.get( KEYS.room(roomId) ); // Call 1  
  room.players.push(player); await redis.set( KEYS.room(room.id),  
  JSON.stringify(room) ); // Call 2 await redis.set( KEYS.playerRoom(player.id),  
  room.id ); // Call 3 } // 3 Redis round trips = latency overhead
```

✅ Solusi - Use Redis Pipeline:

```
✅ BATCHED OPERATIONS:  
  
export async function joinRoom(  
  roomId: string,  
  player: Player,  
  code?: string  
) : Promise {  
  // Read is still individual  
  const raw = await redis.get(KEYS.room(roomId));  
  if (!raw) {  
    return { success: false, error: "Not found" };  
  }  
  
  const room: Room = JSON.parse(raw);  
  
  // Validate...  
  if (room.players.length >= room.maxPlayers) {  
    return { success: false, error: "Room full" };  
  }  
  
  room.players.push(player);  
  
  // ✅ Batch writes in one pipeline  
  const pipeline = redis.pipeline();  
  
  pipeline.set(  
    KEYS.room(room.id),
```

```
    JSON.stringify(room)
  );

  pipeline.set(
    KEYS.playerRoom(player.id),
    room.id
  );

  // ✔ Execute all at once
  await pipeline.exec();

  return { success: true, room };
}

// Result: 1 Redis round trip instead of 2!
// Less latency, better throughput
```

▮ **Alasan:** Reduce Redis round trips, faster response time, better throughput untuk concurrent operations.

Code Review Report - Who Can Make24

Generated: 12 Juni 2026 | Review Scope: Full Project (Client + Server + Shared)

Prepared by: Senior Software Engineer | Next Review: After Critical Issues Fixed